

HTML Conceptual Questions — Answers

1. 200 OK; Content-Type: text/html (normally with a charset such as UTF-8).
2. The HTTP Content-Type header takes precedence because it is supplied by the server before the HTML is parsed. The meta charset is used when the HTTP header does not already specify the encoding.
3. DOCTYPE html tells the browser to use modern standards mode. It is a document type declaration, not an HTML element or tag.
4. A tag is the markup syntax, such as `<p>` or `</p>`. An element is the complete structure, including the opening tag, content, and closing tag when applicable, such as `<p>Hello</p>`.
5. No. Many global attributes are widely valid, but not every attribute is valid on every element without exception. Their allowed use depends on the HTML specification and the element.
6. Generally, inline content should not contain a block-level element. HTML5 made the `<a>` element a special case: it can wrap flow content, including block-level elements, as long as the resulting content is valid.
7. Semantic elements communicate the purpose and structure of content to browsers, search engines, and assistive technologies. A generic `div` with an `id` does not provide the same semantic meaning.
8. `<b>` and `<i>` are mainly for stylistic or alternate-voice purposes, while `<strong>` and `<em>` carry semantic importance or emphasis. Use `<strong>` for important content and `<em>` for emphasis; use `<b>/<i>` when no such semantic meaning is intended.
9. `nbsp` is content, not a layout mechanism. Using it for spacing makes HTML harder to maintain and can cause accessibility and responsive-design problems. CSS margin, padding, and gap should control visual spacing.
10. Use an empty alt attribute: `alt=""`. This tells assistive technologies that the image is decorative and can be ignored. Omitting alt entirely may cause the image filename or other information to be announced.
11. Width and height let the browser reserve the correct amount of space before the image loads. CSS `max-width: 100%` controls how the image fits, but by itself does not reliably reserve its space, so missing dimensions can cause layout shifts.
12. A hero image is normally visible immediately when the page loads. Lazy loading delays its download until it is near the viewport, which can make the main visible content appear later and hurt loading performance.
13. Different source elements let the browser choose a format it supports, such as MP4 or WebM. This improves compatibility across browsers instead of depending on one video format.
14. Historically, `target="_blank"` could allow the newly opened page to access the opener through `window.opener`, creating a reverse-tabnabbing risk. Modern browsers generally provide `noopener` behavior by default, and `rel="noopener"` can also be used explicitly.
15. The label for value must match the input id. This connects the label to the input, so clicking the label focuses or activates the control and assistive technologies can identify its purpose correctly.
16. The input value is not included in the submitted form data because form submission uses the name attribute as the field key. An id identifies the element but does not replace name for submission.

17. `thead`, `tbody`, and `tfoot` communicate the table structure, while `scope='col'` and `scope='row'` identify which headers apply to which cells. This helps browsers and assistive technologies understand relationships in complex tables, not just how the table looks.